

Final Project Rubric with Examples (Individual)

Criteria	Weight	Excellent	Good	Fair / Poor
Problem Formulation & Dataset Choice	15%	Descriptor: Well-scoped question aligned to stakeholder; dataset suitability justified. Example: "Predict hourly energy consumption for smart buildings using sensors (temperature, occupancy, HVAC states); explains relevance for reducing peak loads and sustainability goals."	Descriptor: Mostly clear formulation; partial dataset rationale. Example: "Analyze energy usage trends from sensors" but lacks detail on prediction horizon or stakeholder impact.	Descriptor: Ambiguous or trivial; weak dataset justification. Example: "Analyze building data" without specifying goal or why it matters.
Integration of Skills (Breadth)	25%	Descriptor: Coherent multi-step workflow combining techniques from several courses. Example: Data cleaning (missing IoT readings) → feature engineering (lag variables, weather integration) → model comparison (XGBoost vs LSTM) → optimization (recommend HVAC schedule) → dashboard visualization.	Descriptor: Some integration; missing links. Example: Preprocessing + single model; no comparison or prescriptive step.	Descriptor: Single technique only; no sequential integration. Example: Runs linear regression on raw sensor data without cleaning or additional steps.
Depth of Analysis	35%	Descriptor: Deep, technically sound execution with robust validation. Example: Implements LSTM with hyperparameter tuning, nested cross-validation, error analysis by building type, and sensitivity to weather forecast errors.	Descriptor: Adequate depth; minor validation. Example: Uses random forest with basic k-fold CV; limited error analysis.	Descriptor: Shallow; limited validation/interpretation. Example: Default linear regression with single train-test split; no diagnostics or sensitivity checks.
Recommendations & Impact	10%	Descriptor: Practical actions tied to analysis and stakeholder impact. Example: "Adjust HVAC schedules based on predicted peak hours; expected 12% reduction in energy cost; implement in pilot building cluster."	Descriptor: Plausible but general. Example: "Optimize HVAC usage" without specifying timing or quantified impact.	Descriptor: Unsupported or unrealistic. Example: "Automate building systems" without feasibility or risk discussion."
Report & Code Quality	15%	Descriptor: Professional writing; fully reproducible code. Example: GitHub repo with clear README; clear folder structure; instructions reproduce all figures and results.	Descriptor: Minor gaps in documentation or reproducibility. Example: Code runs with small edits; README lacks environment details.	Descriptor: Disorganized or non-functional. Example: Missing scripts or data paths; no instructions; cannot reproduce results.